

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401289
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Suresh; Anil Cheniveettil et al.

Approach for DC to AC inversion

Abstract

Aspects of the disclosure include a power-conversion system including an input, an output, a DC/DC converter coupled to the input, a DC/AC inverter coupled to the DC/DC converter and coupled to the output, and at least one controller coupled to the DC/DC converter and the DC/AC inverter, the at least one controller being configured to control the DC/DC converter to draw input power from the input and provide converted power to the DC/AC inverter, and control the DC/AC inverter to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input.

Inventors: Suresh; Anil Cheniveettil (Kerala, IN), Jayaraman; Chandrasekaran (Bangalore, IN), Nagabhushanrao; Pradeep Tolakanahalli (Bangalore, IN)

Applicant: SCHNEIDER ELECTRIC IT CORPORATION (Foxboro, MA)

Family ID: 1000008782378

Assignee: SCHNEIDER ELECTRIC IT CORPORATION (Foxboro, MA)

Appl. No.: 17/823676

Filed: August 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230106209 A1	Apr. 06, 2023

Foreign Application Priority Data

IN	202111043888	Sep. 28, 2021
----	--------------	---------------

Publication Classification

Int. Cl.: H02M3/335 (20060101); H02M1/00 (20070101); H02M7/5387 (20070101)

U.S. Cl.:

CPC H02M3/33573 (20210501); H02M1/007 (20210501); H02M7/5387 (20130101);

Field of Classification Search

CPC: H02M (1/007); H02M (3/33573); H02M (7/5387)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6654261	12/2002	Welches et al.	N/A	N/A
7787270	12/2009	NadimpalliRaju et al.	N/A	N/A
8698354	12/2013	Ghosh et al.	N/A	N/A
10277067	12/2018	Mouridsen	N/A	N/A
2010/0133904	12/2009	Klodowski et al.	N/A	N/A
2010/0142236	12/2009	Lee et al.	N/A	N/A
2012/0112547	12/2011	Ghosh	307/66	H02J 9/062
2014/0091718	12/2013	Brinlee	363/16	H05B 45/39
2015/0256111	12/2014	Forster	318/400.27	H02P 6/14
2021/0058004	12/2020	Maki et al.	N/A	N/A
2022/0173652	12/2021	Kusama	N/A	H02M 3/33573

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102007677	12/2010	CN	N/A
2419968	12/2005	GB	N/A
5141594	12/2012	JP	N/A
6190059	12/2016	JP	N/A
201338337	12/2012	TW	N/A
I435526	12/2013	TW	N/A

OTHER PUBLICATIONS

Extended European Search Report from corresponding European Application No. 22197586.5 dated Feb. 13, 2023. cited by applicant

Ramu M. Z. et al.: "A bidirectional inverter with high frequency isolated transformer", Power Engineering Conference, 2003, PECON 2003, Proceedings, National Sangi, Malaysia Dec. 15-16, 2003, Piscataway, NJ, USA, IEEE, Dec. 15, 2003 (Dec. 15, 2003), pp. 71-75, XP010802281, DOI: 10.1109/PECON.2003.1437421 ISBN: 978-0-7803-8208-4. cited by applicant

Primary Examiner: Zhang; Jue

Attorney, Agent or Firm: Lando & Anastasi, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority under 35 U.S.C. § 119 to Indian Application Serial No. 202111043888 titled “A NOVEL APPROACH FOR DC TO AC INVERSION,” filed on Sep. 28, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND

1. Field of the Disclosure

(2) At least one example in accordance with the present disclosure relates generally to power conversion.

2. Discussion of Related Art

(3) Power devices, such as uninterruptible power supplies (UPSs), may include power-conversion components. Power-conversion components may include DC/DC converters and DC/AC inverters. DC/DC converters convert DC power of a first voltage level to DC power of a second voltage level. DC/AC inverters convert DC power to AC power. Some power devices may include a DC/DC converter coupled to, and configured to provide power to, a DC/AC inverter.

SUMMARY

(4) According to at least one aspect of the present disclosure a power-conversion system is provided comprising an input, an output, a DC/DC converter coupled to the input, a DC/AC inverter coupled to the DC/DC converter and coupled to the output, and at least one controller coupled to the DC/DC converter and the DC/AC inverter, the at least one controller being configured to control the DC/DC converter to draw input power from the input and provide converted power to the DC/AC inverter, and control the DC/AC inverter to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input.

(5) In some examples, the DC/DC converter includes converter switches coupled to the input, and the DC/AC inverter includes inverter switches coupled to the output. In at least one example, the at least one controller is further configured to control the converter switches to provide power to the inverter switches in synchronization with controlling the inverter switches to draw power from the converter switches. In various examples, the at least one controller is further configured to control at least two converter switches of the converter switches and at least two inverter switches of the inverter switches to be in a closed and conducting position approximately simultaneously.

(6) In some examples, the DC/DC converter includes a transformer having a primary winding coupled to the converter switches and a secondary winding coupled to the inverter switches. In at least one example, the at least one controller is further configured to control at least two converter switches of the converter switches to conduct power from the input to the primary winding, and control at least two inverter switches of the inverter switches to conduct power from the secondary winding to the output. In various examples, the power-conversion system further includes a positive DC bus and a negative DC bus coupled between the DC/DC converter and the DC/AC inverter; and a capacitor coupled between the positive DC bus and the negative DC bus.

(7) In some examples, the at least one controller is further configured to control the DC/AC inverter to receive the converted power directly from the DC/DC converter, bypassing the capacitor. In at least one example, the capacitor has a capacitance of approximately 50 μF or less. In various examples, the power-conversion system includes an output inductor coupled to the capacitor via the DC/AC inverter and coupled to the output. In some examples, the at least one controller is further configured to control the DC/AC inverter to discharge the output inductor to the capacitor. In at least one example, the at least one controller is further configured to control the DC/AC inverter to discharge the capacitor to the output inductor.

(8) According to at least one aspect, a non-transitory computer-readable medium storing thereon sequences of computer-executable instructions for operating a power-conversion system having an input, an output, a DC/DC converter coupled to the input, and a DC/AC inverter coupled to the DC/DC converter and to the output is provided, the sequences of computer-executable instructions including instructions that instruct at least one processor to control the DC/DC converter to draw input power from the input and provide converted power to the DC/AC inverter, and control the DC/AC inverter to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input.

(9) In some examples, the DC/DC converter includes at least two converter switches and the DC/AC inverter includes at least two inverter switches, and the instructions further instruct at least one processor to control the at least two converter switches to provide power to the at least two inverter switches in synchronization with controlling the at least two inverter switches to draw power from the at least two converter switches. In at least one example, the instructions further instruct at least one processor to control the at least two converter switches and the at least two inverter switches to be in a closed and conducting position approximately simultaneously.

(10) In various examples, the DC/DC converter includes a transformer having a primary winding and a secondary winding, and controlling the at least two converter switches to provide power to the at least two inverter switches includes controlling the at least two converter switches to provide power to the primary winding, and controlling the two or more inverter switches to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input includes controlling the at least two inverter switches to conduct power from the secondary winding to the output.

(11) In some examples, the power-conversion system further includes a positive DC bus and a negative DC bus coupled between the DC/DC converter and the DC/AC inverter, and a capacitor coupled between the positive DC bus and negative DC bus, and the instructions further instruct at least one processor to control the DC/AC inverter to receive the converted power directly from the DC/DC converter, bypassing the capacitor. In at least one example, the power-conversion system further includes a capacitor coupled to the DC/AC inverter and an output inductor coupled to the capacitor via the DC/AC inverter and to the output, and the instructions further instruct at least one processor to control the DC/AC inverter to discharge the output inductor to the capacitor. In various examples, the power-conversion system further includes a capacitor coupled to the DC/AC inverter and an output inductor coupled to the capacitor via the DC/AC inverter and to the output, and the instructions further instruct at least one processor to control the DC/AC inverter to discharge the capacitor to the output inductor.

(12) According to at least one aspect of the disclosure, a method of assembling a power-conversion system is provided, the method comprising providing a power converter having a DC/DC converter, a DC/AC inverter, and at least one controller, coupling the DC/DC converter to the DC/AC inverter, and coupling the at least one controller to the DC/DC converter and to the DC/AC inverter, the at least one controller being configured to control the DC/DC converter to provide power to the DC/AC inverter in synchronization with the DC/AC inverter drawing power from the DC/DC converter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the

limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

- (2) FIG. 1 illustrates a block diagram of a power-conversion system according to an example;
- (3) FIG. 2 illustrates a process of controlling the power-conversion system of FIG. 1 with a non-synchronized control scheme according to an example;
- (4) FIG. 3 illustrates a process of controlling the power-conversion system of FIG. 1 with a synchronized control scheme according to an example;
- (5) FIG. 4 illustrates a schematic diagram of a power-conversion system according to an example;
- (6) FIG. 5 illustrates a process of controlling the power-conversion system of FIG. 4 according to an example;
- (7) FIG. 6 illustrates a graph indicative of a control scheme of the power-conversion system of FIG. 4 according to an example;
- (8) FIG. 7 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a first current according to an example;
- (9) FIG. 8 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a second current according to an example;
- (10) FIG. 9 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a third current according to an example;
- (11) FIG. 10 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a fourth current according to an example;
- (12) FIG. 11 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a fifth current according to an example;
- (13) FIG. 12 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a sixth current according to an example; and
- (14) FIG. 13 illustrates a schematic diagram of the power-conversion system of FIG. 4 including a seventh current according to an example.

DETAILED DESCRIPTION

(15) Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

(16) Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(17) References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for

irreconcilable differences, the term usage in this document controls.

(18) Power devices may include power-conversion devices, such as a DC/DC converter and a DC/AC inverter. The power-conversion device may also include a DC-link capacitor coupled between the DC/DC converter and the DC/AC inverter. The DC-link capacitor may provide an intermediary energy-storage stage to store power provided by the DC/DC converter, and drawn by the DC/AC inverter.

(19) The DC/DC converter may receive input DC power, convert the DC power, and provide converted DC power to the DC-link capacitor to charge the DC-link capacitor. The DC/AC inverter may draw DC power from the DC-link capacitor, invert the DC power, and provide inverted AC power to an output. For example, the DC/AC inverter may provide inverted AC power to a load.

(20) A converter control scheme of the DC/DC converter may be independent from an inverter control scheme of the DC/AC inverter. For example, the DC/DC converter may be controlled to provide power to the DC-link capacitor independently from controlling the DC/AC inverter to draw power from the DC-link capacitor. A control scheme of the DC/DC converter may regard the DC-link capacitor as a load without being aware of the presence of the DC/AC inverter drawing power from the DC-link capacitor. Similarly, a control scheme of the DC/AC inverter may regard the DC-link capacitor as a source without being aware of the presence of the DC/DC converter providing power to the DC-link capacitor.

(21) Accordingly, the DC/DC converter may provide power to the DC-link capacitor independent of an amount of power drawn from the DC-link capacitor by the DC/AC inverter, and the DC/AC inverter may draw power from the DC-link capacitor independent of an amount of power provided to the DC-link capacitor by the DC/DC converter. An amount of power provided by the DC/DC converter to the DC-link capacitor in a certain period of time may therefore differ from an amount of power drawn by the DC/AC converter from the DC-link capacitor in the same period of time.

(22) In some examples, a capacitance of the DC-link capacitor may be relatively high to accommodate filtering requirements and to absorb reactive currents. For example, the DC-link capacitor may be implemented as an electrolytic capacitor. An output inductor may be implemented between the DC/DC converter and the DC-link capacitor to filter power provided by the DC/DC converter to the DC-link capacitor.

(23) In alternate examples, a DC/DC converter may be controlled to provide power to a DC/AC inverter in synchronization with the DC/AC inverter drawing power from the DC/DC converter. "Synchronization" may include the DC/DC converter and DC/AC inverter providing and drawing, respectively, substantially equal amounts of power to one another rather than (or in addition to) providing power to, and drawing power from, a DC-link capacitor as an intermediary stage. In some examples, a capacitor may still be implemented between the DC/DC converter and the DC/AC inverter where the DC/DC converter DC/AC inverter are operated in synchronization. However, in some examples the capacitor in such examples may have a smaller capacitance than in previous examples. For example, the capacitor may be implemented as a film capacitor. Moreover, the filtering output inductor may be omitted in some examples in which power is provided directly from the DC/DC converter to the DC/AC inverter.

(24) Examples of the disclosure provide improved power-conversion devices at least in part by implementing synchronized control of a DC/DC converter and a DC/AC inverter. The DC/DC converter may be controlled to provide power directly to the DC/AC inverter, substantially bypassing a bus capacitor. Providing a synchronized control scheme may simplify control of the DC/DC converter and DC/AC inverter by providing substantially similar or identical control signals to certain components of the DC/DC converter as to certain components of the DC/AC inverter, including converter switches and inverter switches.

(25) Operating the DC/DC converter and DC/AC inverter in synchronization may include closing one or more converter switches approximately simultaneously with one or more inverter switches, as discussed in greater detail below. For example, a duty cycle of the one or more converter

switches and the duty cycle of the one or more synchronized inverter switches may overlap approximately 100% with one another during synchronous operation. As used herein, “approximately simultaneously” may indicate that the duty cycles of the one or more converter switches and the duty cycles of the one or more inverter switches overlap by at least about 99.9% in some examples, by at least about 99% in some examples, by at least about 95% in some examples, by at least about 90% in some examples, or other non-limiting ranges of duty-cycle overlaps. In other examples, “approximately simultaneously” may indicate that control signals are sent to the one or more converter switches within a certain time of sending one or more control signals to the one or more inverter switches, such as within about 0.01 μ s, within about 0.1 μ s, within about 1 μ s, or other non-limiting ranges of time delays.

(26) In examples in which the capacitor is included, the capacitor may be advantageously smaller in capacitance and physical size than in examples in which a DC-link capacitor is implemented as an intermediary stage, and a filtering output inductor may be advantageously omitted. Turn-on losses in switches of the DC/DC converter may be advantageously minimized as explained in greater detail below by implementation of the synchronized control scheme.

(27) Example power-conversion devices may include, or be implemented in, any of several power devices. For purposes of explanation, examples may be provided in which power-conversion devices are implemented in power supplies, such as uninterruptible power supplies (UPSs). In one example, the power-conversion devices include a DC/DC converter and a DC/AC inverter implemented in a power supply that is configured to harness solar power. For example, the solar-power device may receive solar energy, charge an energy-storage device with the solar energy, and discharge the stored energy to the DC/DC converter. The DC/DC converter may convert the received power and provide converted DC power to the DC/AC inverter. The DC/AC inverter may invert the converted DC power and provide AC power to a load, such as a power grid or other load. However, it is to be appreciated that principles of the disclosure are applicable to other types of power devices, other types of energy sources, and other types of loads.

(28) FIG. 1 illustrates a block diagram of a power-conversion system **100** according to an example. The power-conversion system **100** may be implemented in a UPS. For example, the power-conversion system **100** may be implemented in a UPS configured to convert power drawn from solar energy to output power to provide to a load, such as a power grid. The power-conversion system **100** includes a DC input **102**, a DC/DC converter **104**, one or more DC busses **106** (“DC busses **106**”), a DC/AC inverter **108**, an AC output **110**, one or more optional capacitors **112** (“capacitor **112**”), and one or more controllers **114** (“controller **114**”).

(29) The DC input **102** is coupled to the DC/DC converter **104**, and is configured to be coupled to a DC-power source. For example, the DC-power source may include an energy-storage device such as a battery. The DC/DC converter **104** is coupled to the DC input **102** at a first connection, is coupled to the DC busses **106** and the capacitor **112** at a second connection, and is communicatively coupled to the controller **114**. The DC busses **106** are coupled to the DC/DC converter **104** at a first connection and are coupled to the DC/AC inverter **108** at a second connection, and may include or be coupled to the capacitor **112**.

(30) The DC/AC inverter **108** is coupled to the DC busses **106** and the capacitor **112** at a first connection, is coupled to the AC output **110** at a second connection, and is communicatively coupled to the controller **114**. The AC output **110** is coupled to the DC/AC inverter **108**, and is configured to be coupled to a load. For example, the load may be a power grid to which the power-conversion system **100** provides AC power. The capacitor **112** is coupled to the DC busses **106**, the DC/DC converter **104**, and the DC/AC inverter **108**. The controller **114** is communicatively coupled to the DC/DC converter **104** and the DC/AC inverter **108**.

(31) The DC input **102** is configured to receive DC power from a DC-power source, such as a battery, and provide the DC power to the DC/DC converter **104**. The DC/DC converter **104** receives the DC power, converts the DC power to converted DC power, and provides the converted

DC power to the DC busses **106**. Converting the DC power to converted DC power may include stepping up or stepping down a voltage of the DC power received from the DC input **102**. The DC/DC converter **104** is controlled according to one or more control signals provided by the controller **114**.

(32) The DC busses **106** provide the converted DC power to the DC/AC inverter **108** and, in some examples, to the capacitor **112**. The DC busses **106** provide the converted DC power to the DC/AC inverter **108**. The DC/AC inverter **108** receives the converted DC power, inverts the converted DC power to inverted AC power, and provides the inverted AC power to the AC output **110**. The DC/AC inverter **108** is controlled according to one or more control signals provided by the controller **114**.

(33) The controller **114** is configured to control operation of the power-conversion system **100**, such as by controlling operation of the DC/DC converter **104** and the DC/AC inverter **108**. The DC/DC converter **104** and the DC/AC inverter **108** may each include one or more switches. The controller **114** may control the switches by providing control signals, such as pulse-width-modulation (PWM) signals, to the switches. In some examples, the controller **114** may synchronize control of the DC/DC converter **104** with control of the DC/AC inverter **108**. In other examples, the controller **114** may not synchronize control of the DC/DC converter **104** with control of the DC/AC inverter **108**. An example of a non-synchronized control scheme executed by the controller **114** is provided with respect to FIG. 2. An example of a synchronized control scheme executed by the controller **114** is provided with respect to FIG. 3.

(34) FIG. 2 illustrates a process **200** of controlling the power-conversion system **100** with a non-synchronized control scheme according to an example. The process **200** may be executed at least in part by the controller **114**, the DC/DC converter **104**, and the DC/AC inverter **108**. It is to be appreciated that acts of the process **200** are illustrated sequentially for purposes of explanation only. In some examples, one or more acts of the process **200** may be executed substantially simultaneously. Furthermore, the order of acts of the process **200** are provided for purposes of explanation only, and in some examples acts of the process **200** may be executed in a different order.

(35) At act **202**, the process **200** begins.

(36) At act **204**, the controller **114** controls the DC/DC converter **104** to convert DC power received from the DC input **102**. Controlling the DC/DC converter **104** may include providing control signals to one or more switches in the DC/DC converter **104** to step-up or step-down a voltage level of the DC power received from the DC input **102**. In some examples, the DC/DC converter **104** includes a step-up transformer, and act **204** includes controlling the one or more switches to provide power from the DC input **102** to a primary winding of the step-up transformer. Converted DC power may be output at a secondary winding of the step-up transformer. Controlling the one or more switches in the DC/DC converter **104** may include alternately coupling the DC input **102** to a first connection of the transformer and a second connection of the transformer.

(37) At act **206**, converted DC power is provided from the DC/DC converter **104** to the capacitor **112**. Continuing with the example above, the secondary winding of the step-up transformer may provide the converted DC power to the capacitor **112** to charge the capacitor **112**.

(38) At act **208**, the controller **114** controls the DC/AC inverter **108** to draw power from the capacitor **112**. Controlling the DC/AC inverter **108** may include providing control signals to one or more switches in the DC/AC inverter **108** to draw power from the capacitor **112**. An amount of power drawn from the capacitor **112** at act **208** may not be equal to an amount of power provided to the capacitor **112** at act **204** in examples in which the controller **114** does not synchronize control of the DC/DC converter **104** at act **204** with control of the DC/AC inverter **108** at act **208**.

(39) At act **210**, the controller **114** controls the DC/AC inverter **108** to convert the power drawn from the capacitor **112** to AC power. As discussed above, the DC/AC inverter **108** may include one or more inverter switches. Act **210** may include providing one or more PWM signals to the one or

more inverter switches to invert the DC power and produce inverted AC power.

(40) At act **212**, the DC/AC inverter **108** provides the inverted AC power to the AC output **110**. The inverted AC power may be provided from the AC output **110** to a load. For example, as discussed above, the load may include an AC-power grid. The process **200** then returns to act **204**. As discussed above, acts of the process **200** may be executed in a different order, and acts of the process **200** may be executed substantially simultaneously. Similarly, the process **200** may proceed from act **212** to any of acts **204-210**.

(41) Accordingly, the process **200** may provide an example of a non-synchronized control scheme at least because the controller **114** controls the DC/DC converter **104** independently from the DC/AC inverter **108**. The controller **114** controls the DC/DC converter **104** to provide power to the capacitor **112** and separately controls the DC/AC inverter **108** to draw power from the capacitor **112**. An amount of power provided to the capacitor **112** may differ from an amount of power drawn from the capacitor **112** in a non-synchronized control scheme.

(42) FIG. 3 illustrates a process **300** of controlling the power-conversion system **100** with a synchronized control scheme according to another example. The process **300** may be executed at least in part by the controller **114**, the DC/DC converter **104**, and the DC/AC inverter **108**. It is to be appreciated that acts of the process **300** are illustrated sequentially for purposes of explanation only. In some examples, one or more acts of the process **300** may be executed substantially simultaneously. Furthermore, the order of acts of the process **300** are provided for purposes of explanation only, and in some examples acts of the process **300** may be executed in a different order.

(43) At act **302**, the process **300** begins.

(44) At act **304**, the controller **114** controls the DC/DC converter **104** to convert DC power in synchronization with the DC/AC inverter **108** inverting power. As discussed above, the DC/DC converter **104** may include one or more switches and at least one step-up transformer. The controller **114** may provide PWM signals to one or more switches of the DC/DC converter **104** and the DC/AC inverter **108** such that periods of time in which the DC/DC converter **104** is converting DC power to provide to the DC/AC inverter **108** is aligned with (for example, synchronized with) periods of time in which the DC/AC inverter **108** is inverting DC power to produce AC power.

(45) At act **306**, the DC/DC converter **104** provides converted DC power to the DC/AC inverter **108**. The DC/DC converter **104** may provide the converted DC power directly to the DC/AC inverter **108**, substantially bypassing the capacitor **112**. In some examples, a portion of the converted DC power may be provided to the capacitor **112**. However, a substantial portion of the converted DC power may be provided directly to the DC/AC inverter **108** from the DC/DC converter **104** without first being provided to the capacitor **112**.

(46) At act **308**, the controller **114** controls the DC/AC inverter **108** to convert DC power to AC power in synchronization with the DC/DC converter **104** converting power. The DC/AC inverter **108** may include one or more inverter switches. As discussed above, the controller **114** may control the one or more inverter switches to draw power from the DC/DC converter **104** in synchronization with the DC/DC converter **104** providing power to the DC/AC inverter **108**, such that the DC/AC inverter **108** draws power substantially directly from the DC/DC converter **104** rather than substantially from the capacitor **112**.

(47) At act **310**, the DC/AC inverter **108** provides inverted AC power to the AC output **110**. The inverted AC power may be provided from the AC output **110** to a load. For example, as discussed above, the load may include an AC-power grid. The process **300** then returns to act **304**. As discussed above, acts of the process **300** may be executed in a different order, and acts of the process **300** may be executed substantially simultaneously. Similarly, the process **300** may proceed from act **310** to any of acts **304-308**.

(48) Accordingly, the process **300** may provide an example of a synchronized control scheme at least because the controller **114** substantially aligns (for example, synchronizes) periods of time

during which power is provided by the DC/DC converter **104** with periods of time during which power is drawn by the DC/AC inverter **108**. Power provided by the DC/DC converter **104** to the DC/AC inverter **108** may substantially bypass the capacitor **112**. An amount of power provided by the DC/DC converter **104** may be substantially equal to an amount of power drawn by the DC/AC inverter **108** in a given period of time during synchronized control.

(49) FIG. **4** illustrates a schematic diagram of a power-conversion system **400** according to an example. The power-conversion system **400** may illustrate one example of the power-conversion system **100**, and similar components are labeled accordingly. The power-conversion system **400** includes the DC input **102**, the DC/DC converter **104**, the DC busses **106**, a DC/AC inverter **108**, the AC output **110**, the capacitor **112**, and the controller **114**. The DC input **102** includes a first DC input **102a** and a second DC input **102b**. The DC busses **106** include a first DC bus **106a** (or a “positive DC bus”) and a second DC bus **106b** (or a “negative DC bus”). The AC output **110** includes a first AC output **110a** and a second AC output **110b**. The power-conversion system **400** is configured to be coupled to an energy-storage device **402** via the DC input **102**, and is configured to be coupled to a load **404** via the AC output **110**.

(50) The DC/DC converter **104** includes a first converter switch **406**, a second converter switch **408**, a third converter switch **410**, a fourth converter switch **412**, a transformer **414** including a primary winding **416** and a secondary winding **418**, a first diode **420**, a second diode **422**, a third diode **424**, and a fourth diode **426**. The DC/AC inverter **108** includes a first inverter switch **428**, a second inverter switch **430**, a third inverter switch **432**, a fourth inverter switch **434**, an inductor **436** (or “output inductor **436**”), and a capacitor **438**.

(51) The first DC input **102a** is coupled to the first converter switch **406** and the third converter switch **410**, and is configured to be coupled to the energy-storage device **402**. The second DC input **102b** is coupled to the second converter switch **408** and to the fourth converter switch **412**, and is configured to be coupled to the energy-storage device **402**. In one example, the energy-storage device **402** includes a positive terminal configured to be coupled to the first DC input **102a**, and a negative terminal configured to be coupled to the second DC input **102b**.

(52) The first converter switch **406** is coupled to the first DC input **102a** and the third converter switch **410** at a first connection, is coupled to the second converter switch **408** and the primary winding **416** at a second connection, and is communicatively coupled to the controller **114** at a control connection. The second converter switch **408** is coupled to the first converter switch **406** and the primary winding **416** at a first connection, is coupled to the second DC input **102b** and the fourth converter switch **412** at a second connection, and is communicatively coupled to the controller **114** at a control connection.

(53) The third converter switch **410** is coupled to the first DC input **102a** and the first converter switch **406** at a first connection, is coupled to the fourth converter switch **412** and the primary winding **416** at a second connection, and is communicatively coupled to the controller **114** at a control connection. The fourth converter switch **412** is coupled to the third converter switch **410** and the primary winding **416** at a first connection, is coupled to the second DC input **102b** and the second converter switch **408** at a second connection, and is communicatively coupled to the controller **114** at a control connection.

(54) The primary winding **416** is coupled to the third converter switch **410** and the fourth converter switch **412** at a first connection, is coupled to the first converter switch **406** and the second converter switch **408** at a second connection, and is magnetically coupled to the secondary winding **418**. The secondary winding **418** is coupled to the first diode **420** and the second diode **422** at a first connection, is coupled to the third diode **424** and the fourth diode **426** at a second connection, and is magnetically coupled to the primary winding **416**.

(55) The first diode **420** is coupled to the third diode **424**, the capacitor **112**, the first inverter switch **428**, and the third inverter switch **432** via the first DC bus **106a** at a cathode connection, and is coupled to the secondary winding **418** and the second diode **422** at an anode connection. The

second diode **422** is coupled to the first diode **420** and the secondary winding **418** at a cathode connection, and is coupled to the fourth diode **426**, the capacitor **112**, the second inverter switch **430**, and the fourth inverter switch **434** via the second DC bus **106b** at an anode connection.

(56) The third diode **424** is coupled to the first diode **420**, the capacitor **112**, the first inverter switch **428**, and the third inverter switch **432** via the first DC bus **106a** at a cathode connection, and is coupled to the secondary winding **418** and the fourth diode **426** at an anode connection. The fourth diode **426** is coupled to the third diode **424** and the secondary winding **418** at a cathode connection, and is coupled to the second diode **422**, the capacitor **112**, the second inverter switch **430**, and the fourth inverter switch **434** via the second DC bus **106b** at an anode connection.

(57) The capacitor **112** is coupled to the first diode **420**, the third diode **424**, the first inverter switch **428**, and the third inverter switch **432** at a first connection via the first DC bus **106a**, and is coupled to the second diode **422**, the fourth diode **426**, the second inverter switch **430**, and the fourth inverter switch **434** at a second connection via the second DC bus **106b**. The first DC bus **106a** is coupled to the first diode **420**, the third diode **424**, the first inverter switch **428**, and the third inverter switch **432**. The second DC bus **106b** is coupled to the second diode **422**, the fourth diode **426**, the second inverter switch **430**, and the fourth inverter switch **434**.

(58) The first inverter switch **428** is coupled to the first diode **420**, the third diode **424**, the capacitor **112**, and the third inverter switch **432** via the first DC bus **106a**, is coupled to the second inverter switch **430**, the capacitor **438**, and the second AC output **110b** at a second connection, and is communicatively coupled to the controller **114** at a control connection. The second inverter switch **430** is coupled to the first inverter switch **428**, the capacitor **438**, and the second AC output **110b** at a first connection, is coupled to the second diode **422**, the fourth diode **426**, the capacitor **112**, and the fourth inverter switch **434** via the second DC bus **106b** at a second connection, and is communicatively coupled to the controller **114** at a control connection.

(59) The third inverter switch **432** is coupled to the first diode **420**, the third diode **424**, the capacitor **112**, and the first inverter switch **428** via the first DC bus **106a** at a first connection, is coupled to the inductor **436** and the fourth inverter switch **434** at a second connection, and is communicatively coupled to the controller **114** at a control connection. The fourth inverter switch **434** is coupled to the third inverter switch **432** and the inductor **436** at a first connection, is coupled to the second diode **422**, the fourth diode **426**, the capacitor **112**, and the second inverter switch **430** at a second connection, and is communicatively coupled to the controller **114** at a control connection.

(60) The inductor **436** is coupled to the capacitor **438** and the first AC output **110a** at a first connection, and is coupled to the third inverter switch **432** and the fourth inverter switch **434** at a second connection. The capacitor **438** is coupled to the first inverter switch **428**, the second inverter switch **430**, and the second AC output **110b** at a first connection, and is coupled to the inductor **436** and the first AC output **110a** at a second connection. The first AC output **110a** is coupled to the inductor **436** and the capacitor **438**, and is configured to be coupled to the load **404**. The second AC output **110b** is coupled to the first inverter switch **428**, the second inverter switch **430**, and the capacitor **438**, and is configured to be coupled to the load **404**. The load **404** is configured to be coupled to the first AC output **110a** at a first connection and is configured to be coupled to the second AC output **110b** at a second connection.

(61) The controller **114** is communicatively coupled to the converter switches **406-412** and to the inverter switches **428-434**. The controller **114** is configured to provide one or more control signals to the converter switches **406-412** and to the inverter switches **428-434**. For example, the controller **114** may provide one or more PWM signals to the converter switches **406-412** and/or to the inverter switches **428-434** as discussed above with respect to the process **300**. An example of controlling the power-conversion system **400** is provided with respect to FIG. 5.

(62) FIG. 5 illustrates a process **500** of controlling the power-conversion system **400** according to an example. The process **500** may be executed at least in part by the controller **114**. Acts of the

process **500** may be executed simultaneously, and may be executed in a different order than that illustrated in the following examples. Relatedly, FIG. **6** illustrates a graph **600** indicative of a control scheme executed by the controller **114** according to an example. As discussed below, the graph **600** is indicative of control signals provided by the controller **114** pursuant to the acts of the process **500**.

(63) At act **502**, the process **500** begins.

(64) At act **504**, a determination is made as to whether to control the DC/DC converter **104** to draw DC power from the energy-storage device **402**. Power may be drawn from the energy-storage device **402** based on a control scheme executed by the controller **114**. In examples in which the power-conversion system **400** outputs an AC waveform at the AC output **110**, power may be drawn from the energy-storage device **402** according to a target amplitude of the output AC waveform. For example, more power may be drawn from the energy-storage device **402** while outputting high-amplitude portions of the AC waveform compared to outputting lower-amplitude portions of the AC waveform. Examples are provided with respect to the graph **600** to illustrate the foregoing.

(65) The graph **600** includes a carrier-waveform trace **602**, a first reference-waveform trace **604**, a second reference-waveform trace **606**, a first control-signal trace **608** indicative of a control signal provided to the first inverter switch **428**, a second control-signal trace **610** indicative of a control signal provided to the third inverter switch **432**, a third control-signal trace **612** indicative of a second inverter switch **430**, a fourth control-signal trace **614** indicative of a control signal provided to the fourth inverter switch **434**, a fifth control-signal trace **616** indicative of a control signal provided to the first converter switch **406**, a sixth control-signal trace **618** indicative of a control signal provided to the second converter switch **408**, a seventh control-signal trace **620** indicative of a control signal provided to the third converter switch **410**, and an eighth control-signal trace **622** indicative of a control signal provided to the fourth converter switch **412**.

(66) The traces **602-606** provide a visual indication of a determination of when to draw power from the energy-storage device **402**. The reference-waveform traces **604**, **606** indicate a target output AC waveform and an inversion thereof. In one example, the controller **114** determines to control the converter switches **406-412** to draw power from the energy-storage device **402** where the value of the carrier-waveform trace **602** is between the values of the reference-waveform traces **604**, **606**.

(67) During these time periods, the primary winding **416** may draw power from the energy-storage device **402** by the controller **114** closing two of the converter switches **406-412**, such as the first converter switch **406** and the fourth converter switch **412** or the second converter switch **408** and the third converter switch **410**. With reference to the graph **600**, power may be drawn from the energy-storage device **402** where the sixth control-signal trace **618** and seventh control-signal trace **620** are simultaneously high, corresponding to closing the converter switches **408**, **410**, or where the fifth control-signal trace **616** and eighth control-signal trace **622** are simultaneously high, corresponding to closing the converter switches **406**, **412**. Accordingly, the traces **618**, **620** or the traces **616**, **622** may be simultaneously high where a value of the carrier-waveform trace **602** is between the values of the reference-waveform traces **604**, **606**.

(68) For example, between a first time **624** and a second time **626**, the value of the carrier-waveform trace **602** is between the values of the reference-waveform traces **604**, **606**, and the controller **114** may determine that power should be drawn from the energy-storage device **402**. Accordingly, act **504** may be visually represented by determining whether the carrier-waveform trace **602** is between the reference-waveform traces **604**, **606**. Responsive to determining that DC power is to be drawn from the energy-storage device **402** (**504 YES**), the process **500** continues to act **506**.

(69) At act **506**, the controller **114** controls the DC/DC converter **104** to draw DC power from the energy-storage device **402**. As discussed above, controlling the DC/DC converter **104** to draw DC power from the energy-storage device **402** may include controlling two of the converter switches **406-412** to couple the energy-storage device **402** to the primary winding **416** of the transformer

414. For example, between the first time **624** and the second time **626**, the controller **114** may provide control signals to close the second converter switch **408** and the third converter switch **410** as indicated by the sixth control-signal trace **618** and the seventh control-signal trace **620** being simultaneously high.

(70) It is to be appreciated that the period of time between the first time **624** and the second time **626** is only one example of the DC/DC converter **104** drawing DC power from the energy-storage device **402**. With reference to the graph **600**, the DC/DC converter **104** may draw DC power from the energy-storage device **402** where the traces **618**, **620** or the traces **616**, **622** are simultaneously high. Accordingly, while examples of act **506** include controlling the converter switches **408**, **410** to couple the primary winding **416** to the energy-storage device **402**, other examples of act **506** may include controlling the converter switches **406**, **412** to couple the primary winding **416** to the energy-storage device **402**.

(71) Closing the converter switches **408**, **410** electrically couples the primary winding **416** to the energy-storage device **402**. FIG. 7 illustrates a schematic diagram of the power-conversion system **400** in which the converter switches **408**, **410** are closed and conductive, thereby coupling the primary winding **416** to the energy-storage device **402**. A current path **700** indicates a path of current provided to the primary winding **416** by the energy-storage device **402**. As indicated by the current path **700**, a current is received from the energy-storage device **402** at the first DC input **102a**, and passes through the third converter switch **410**, the primary winding **416**, and the second converter switch **408**, and returns to the energy-storage device **402** via the second DC input **102b**. A current passing through the primary winding **416** may induce a voltage and/or current in the secondary winding **418** of the transformer **414**.

(72) At act **508**, a determination is made as to whether a voltage of the secondary winding **418** is greater than a voltage of the capacitor **112**. Providing power to the primary winding **416** of the transformer **414** may induce a voltage across the secondary winding **418** of the transformer **414** as discussed above. However, the voltage across the secondary winding **418** may not be greater than a voltage across the capacitor **112**. Because the secondary winding **418** is coupled to, and configured to discharge power through, either the first diode **420** or third diode **424**, the secondary winding **418** may not discharge power to the first DC bus **106a** if the voltage of the secondary winding **418** is less than the voltage of the capacitor **112**, because the diodes **420**, **424** may be reverse-biased. In at least one example, act **508** does not include any particular component performing a determination, and act **508** is illustrated for purposes of explanation to discuss a situation in which the diodes **420**, **424** are reverse-biased, and a situation in which the diodes **420**, **424** are forward-biased.

(73) For example, if a voltage is high at the dotted connection of the secondary winding **418** but is less than a voltage of the capacitor **112** (**508 NO**), the first diode **420** will be reverse-biased and will not be conductive. Otherwise, if a voltage is high at the dotted connection of the secondary winding **418** and is greater than a voltage of the capacitor **112** (**508 YES**), the first diode **420** will be forward-biased and will be conductive, and the process **500** continues to act **510**.

(74) Similarly, if a voltage is high at the non-dotted connection of the secondary winding **418** but is less than a voltage of the capacitor **112** (**508 NO**), the third diode **424** will be reverse-biased and will not be conductive. Otherwise, if a voltage is high at the non-dotted connection of the secondary winding **418** and is greater than a voltage of the capacitor **112** (**508 YES**), the third diode **424** will be forward-biased and will be conductive, and the process **500** continues to act **510**. A situation in which the first diode **420** or third diode **424** is forward-biased will first be discussed.

(75) At act **510**, the DC/DC converter **104** provides DC power to the first DC bus **106a**. DC power may be provided to the first DC bus **106a** via either the first diode **420** or third diode **424**, depending on whether a dotted or non-dotted connection of the secondary winding **418** is high. FIG. 8 illustrates a schematic diagram of the power-conversion system **400** in which the dotted connection of the secondary winding **418** is high and, as discussed above, a voltage of the

secondary winding **418** is greater than a voltage of the capacitor **112**. At least because the first diode **420** is forward-biased, the secondary winding **418** discharges current to the first DC bus **106a** via the first diode **420**. A current path **800** indicates a path of current discharged by the secondary winding **418**. As indicated by the current path **800**, the current passes from the secondary winding **418** through the first diode **420** to the first DC bus **106a**.

(76) At act **512**, the controller **114** controls the DC/AC inverter **108** to generate an AC output using power provided to the first DC bus **106a** from the DC/DC converter **104**. Controlling the DC/AC inverter **108** to generate an AC output may include simultaneously closing the inverter switches **428**, **434** or the inverter switches **430**, **432** at varying pulse widths, as indicated by periods of time at which the control-signal traces **608**, **614** or control-signal traces **610**, **612** are simultaneously high. For example, the inverter switches **428**, **434** may be intermittently closed during negative half-cycles of an output AC waveform, and the inverter switches **430**, **432** may be intermittently closed during positive half-cycles of an output AC waveform.

(77) FIG. **8** illustrates an example of a time period during which the control-signal traces **610**, **612** are simultaneously high, such as between the first time **624** and the second time **626**, which may correspond to a positive half-cycle of an output AC waveform. A first portion of the current indicated by the current path **800** passes through the capacitor **112** (for example, to charge the capacitor **112**) and returns to the secondary winding **418** via the fourth diode **426**.

(78) A second portion of the current is provided directly to the DC/AC inverter **108**, which the DC/AC inverter **108** may use to generate an AC output by controlling the inverter switches **430**, **432** to intermittently open and close. The second portion of the current may bypass the capacitor **112**. Bypassing the capacitor may include the DC/DC converter **104** providing the second portion of the current directly to the DC/AC inverter **108** without first being provided to the capacitor **112**. The inverter switches **430**, **432** may be closed substantially simultaneously with the second converter switch **408** and third inverter switch **410** between the first time **624** and second time **626**, that is, in synchronization with the converter switches **408**, **410**.

(79) Similarly, the inverter switches **428**, **434** may be intermittently closed during a negative half-cycle of an output AC waveform as indicated by the traces **608**, **614**. For example, between a third time **628** and a fourth time **630**, the inverter switches **428**, **434** may be simultaneously closed. As indicated by the traces **616**, **622**, the converter switches **406**, **412** may be closed in synchronization with the inverter switches **428**, **434** between the third time **629** and fourth time **630**.

(80) At act **514**, the DC/AC inverter **108** provides an AC output waveform generated by the inverter switches **428-434** to the AC output **110**. Returning to FIG. **8**, the current path **800** may indicate an example of current between the first time **624** and the second time **626** during which the second inverter switch **430** and third inverter switch **432** are closed. The current path **800** passes through the third inverter switch **432**, the inductor **436**, the first AC output **110a**, the load **404**, the second AC output **110b**, and the second inverter switch **430**, and returns to the secondary winding **418** via the fourth diode **426**. A portion of the current may bypass the load **404** and be filtered via the capacitor **438**. The process **500** then returns to act **504**.

(81) In the foregoing examples, the current path **800** may indicate a path of current for which a voltage across the inductor **436** is in phase with a current through the inductor **436**. As appreciated by one of ordinary skill in the art, a voltage across an inductor may not always be in phase with a current through an inductor, such as in examples in which the load **404** is a reactive load. Accordingly, although a voltage may be dropped across the inductor **436** from the first DC bus **106a** to the second DC bus **106b**, a current may be discharged by the inductor **436** to the first DC bus **106a** where an inductor current is out of phase with an inductor voltage.

(82) FIG. **9** illustrates a schematic diagram of the power-conversion system **400** that is similar to the schematic diagram of FIG. **8**, but in which a current through the inductor **436** is out of phase with a voltage across the inductor **436**. A first current path **900** indicates a path of current discharged by the inductor **436** while the converter switches **408**, **410** and inverter switches **430**,

432 are closed, such as between the first time **624** and second time **626**. As indicated by the current path **900**, the inductor **436** discharges current through the third inverter switch **432** to the capacitor **112** via the first DC bus **106a**, such that the capacitor **112** absorbs reactive power stored in the inductor **436**. The current returns to the inductor **436** by passing through the second inverter switch **430**, the second AC output **110b**, the load **404**, and the first AC output **110a** to the inductor **436**. A second current path **902** indicates a path of current discharged by the secondary winding **418** through the first diode **420**, the capacitor **112**, the fourth diode **426**, and back to the secondary winding **418**.

(83) Accordingly, while the switches **408**, **410**, **430**, **432** are closed and a voltage of the secondary winding **418** exceeds a voltage of the capacitor **112**, current through the inductor **436** may follow either the current path **800** or the first current path **900** depending on a phase relationship between the current and voltage of the inductor **436** at a particular time.

(84) Returning to act **508**, if a voltage of the secondary winding **418** is not greater than a voltage of the capacitor **112** (**508 NO**), then the process **500** continues to act **516**.

(85) At act **516**, power is exchanged between the capacitor **112** and the DC/AC inverter **108**. At least because the first diode **420** and/or third diode **424** are reverse-biased, the DC/DC converter **104** may not provide power to the DC/AC inverter **108**. However, the capacitor **112** may discharge stored energy to the DC/AC inverter **108**, and/or the inductor **436** may discharge stored energy to the capacitor **112**.

(86) Act **516** may be executed between the first time **624** and the second time **626**, for example. The inverter switches **430**, **432** are closed to couple the DC/AC inverter **108** to the capacitor **112**. The capacitor **112** may provide a voltage to the DC/AC inverter **108**. Act **516** may include the capacitor **112** discharging stored power to the load **404** where a voltage across the inductor **436** is in phase with a current through the inductor **436**, or may include the inductor **436** discharging stored power to the capacitor **112** where a voltage across the inductor **436** is not in phase with the current through the inductor **436**.

(87) FIG. **10** illustrates a schematic diagram of the power-conversion system **400** in which a voltage across the inductor **436** is in phase with a current through the inductor **436**, and the capacitor **112** discharges stored power to the inductor **436**. A current path **1000** indicates a path of current discharged by the capacitor **112**. As indicated by the current path **1000**, a current discharged by the capacitor **112** passes through the third inverter switch **432**, the inductor **436**, the first AC output **110a**, the load **404**, the second AC output **110b**, and the second inverter switch **430**, and returns to the capacitor **112**. The process **500** continues to act **512**, and the DC/AC inverter **108** generates an AC output based on the power provided to the AC output **110** as discussed above.

(88) FIG. **11** illustrates a schematic diagram of the power-conversion system **400** in which a voltage across the inductor **436** is not in phase with a current through the inductor **436**, and the inductor **436** therefore discharges stored power to the capacitor **112**. A current path **1100** indicates a path of current discharged by the inductor **436**. As indicated by the current path **1100**, a current discharged by the inductor **436** passes through the third inverter switch **432**, the capacitor **112**, the second inverter switch **430**, the second AC output **110b**, the load **404**, and the first AC output **110a**, and returns to the inductor **436**. The process **500** continues to act **512**, and the DC/AC inverter **108** generates an AC output based on the power provided to the AC output **110** as discussed above.

(89) Returning to act **504**, if a determination is made to not draw DC power from the energy-storage device **402** (**504 NO**), then the process **500** continues to act **518**. In various examples, power may be drawn from the energy-storage device **402** when the DC/AC inverter **108** is drawing power from the DC/DC converter **104**. For example, the DC/AC inverter **108** may draw more power from the DC/DC converter **104** during higher-amplitude portions of an output AC waveform. When the DC/AC inverter **108** is not drawing power from the DC/DC converter **104**, the DC/DC converter **104** may be controlled to not draw power from the energy-storage device **402** at least because the DC/DC converter **104** is synchronized with the DC/AC inverter **108**. In various

examples, the controller **114** may instead control the DC/DC converter **104** to discharge energy stored in the transformer **414** while not drawing power from the energy-storage device **402**.

(90) At act **518**, the controller **114** controls the DC/DC converter **104** to discharge energy stored in the primary winding **416**. Act **518** may include decoupling the primary winding **416** from the energy-storage device **402** and shorting the primary winding **416**. More particularly, act **518** may include simultaneously closing the second converter switch **408** and the fourth inverter switch **412** or simultaneously closing the first converter switch **406** and the third converter switch **410**. For example, between a fifth time **632** and a sixth time **634**, the controller **114** may simultaneously close the second converter switch **408** and the fourth inverter switch **412**, as indicated by the sixth control-signal trace **618** and the eighth control-signal trace **622**.

(91) FIG. **12** illustrates a schematic diagram of the power-conversion system **400** in which the second converter switch **408**, the fourth converter switch **412**, the second inverter switch **430**, and the fourth inverter switch **434** are closed, such as between the fifth time **632** and the sixth time **634**. A first current path **1200** indicates a path of current discharged by the primary winding **416**. As indicated by the current path **1200**, the current is discharged by the primary winding **416** through the second converter switch **408** and the fourth converter switch **412**, and returns to the primary winding **416**. In other examples, act **518** may include simultaneously closing the converter switches **406**, **410** to discharge the primary winding **416**.

(92) At act **520**, the controller **114** controls the DC/AC inverter **108** to discharge energy stored in the inductor **436**. Act **520** may include decoupling the DC/AC inverter **108** from the DC/DC converter **104** and coupling the inductor **436** to the load **404**. More particularly, act **520** may include simultaneously closing the first inverter switch **428** and the third inverter switch **432** or simultaneously closing the second inverter switch **430** and the fourth inverter switch **434**. For example, between the fifth time **632** and the sixth time **634**, the controller **114** may simultaneously close the second inverter switch **430** and the fourth inverter switch **434**, as indicated by the third control-signal trace **612** and the fourth control-signal trace **614**.

(93) FIG. **12** includes a second current path **1202** indicating a path of current discharged by the inductor **436** in a first example. As indicated by the current path **1202**, the current is discharged by the inductor **436** to the first AC output **110a**, the load **404**, the second AC output **110b**, the second inverter switch **430**, and the fourth inverter switch **434**. As discussed above, the inductor **436** may discharge bidirectionally and may therefore discharge current in an opposite direction of the first example of FIG. **12**.

(94) FIG. **13** illustrates a schematic diagram of the power-conversion system **400** in which the second converter switch **408**, the fourth converter switch **412**, the second inverter switch **430**, and the fourth inverter switch **434** are closed, such as between the second time **626** and the fifth time **632**. The current path **1200** indicates a path of a current discharged by the primary winding **416**, as discussed above. A current path **1300** indicates a path of current discharged by the inductor **436** according to a second example. As indicated by the current path **1300**, the current is discharged from the inductor **436** to the fourth inverter switch **434**, the second inverter switch **430**, the second AC output **110b**, the load **404**, and the first AC output **110a**, and returns to the inductor **436**. The process **500** then returns to act **504**.

(95) Accordingly, FIG. **6** provides a graph **600** illustrating a control scheme and control signals implemented by the controller **114**, and FIG. **5** provides a process **500** indicative of the control scheme. Control of the DC/DC converter **104** is synchronized with control of the DC/AC inverter **108** at least in part by aligning control signals of the converter switches **406-412** with control signals of the inverter switches **428-434**. For example, PWM signals are provided to the converter switches **406-412** as indicated by the traces **616-622**, and the PWM signals are synchronized with the PWM signals provided to the inverter switches **428-434** as indicated by the traces **608-614**.

(96) As discussed above, the secondary winding **418** may provide power to the capacitor **112** to charge the capacitor **112**. The secondary winding **418** may therefore stop charging the capacitor **112**

when the capacitor **112** is charged to approximately a voltage of the secondary winding **418**. However, the inductor **436** may discharge reactive power to the capacitor **112** as discussed above, which may charge the capacitor **112** to a voltage above a voltage of the secondary winding **418** as discussed above with respect to act **508**. The capacitor **112** may later discharge the stored power to the inductor **436**.

(97) In some examples, the capacitor **112** has a capacitance of approximately 50 μ F. As used herein, “approximately 50 μ F” may include a capacitance between, for example, about 49.9 and about 50.1 μ F, between about 49 and about 51 μ F, between about 45 and about 55 μ F, between about 40 and about 60 μ F, or other non-limiting examples. In some examples, the capacitor **112** is physically smaller and has a smaller capacitance than an existing DC-link capacitor. For example, whereas a DC-link capacitor may be implemented as an electrolytic capacitor, the capacitor **112** may be implemented as a film capacitor (for example, having a capacitance of approximately 50 μ F) in some examples. In still other examples, the capacitor **112** may be omitted.

(98) In some examples, the power-conversion systems **100**, **400** are, or are implemented in, a UPS. In one example, the UPS may include a 230 V, 1000 VA UPS operating at a switching frequency of approximately 30 kHz and having a 230 V, 50 Hz AC output. The DC input **102** may be coupled to an energy-storage device, such as the energy-storage device **402**, of various voltages. In one example, the energy-storage device **402** includes a 48 V battery. The transformer **414** may include a step-down or step-up transformer. In one example, the transformer **414** has a turns ratio of 2:17, is configured to receive an input voltage of approximately 48 V from the energy-storage device **402**, and is configured to provide a voltage of approximately 400 V across the DC busses **106**. The AC output **110** is configured to provide an output signal which may have any of several voltages or frequencies, such as a 230 V, 50 Hz output signal. In other examples, the power-conversion systems **100**, **400** may be, or may be implemented in, power devices other than UPSs and/or UPS s having different operating parameters.

(99) Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of, and within the spirit and scope of, this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A power-conversion system comprising: an input; an output; a DC/DC converter coupled to the input, wherein the DC/DC converter includes converter switches coupled to the input; a DC/AC inverter coupled to the DC/DC converter and coupled to the output, wherein the DC/AC inverter includes inverter switches coupled to the output; and at least one controller coupled to the DC/DC converter and the DC/AC inverter, the at least one controller being configured to control the DC/DC converter based on a carrier signal having a carrier-signal frequency to draw input power from the input and provide converted power to the DC/AC inverter, control the DC/AC inverter to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input, wherein controlling the DC/AC inverter to provide the output power to the output in synchronization with the DC/DC converter drawing the input power from the input includes controlling the DC/AC inverter based on the carrier signal having the carrier-signal frequency, control the DC/DC converter switches based on a comparison between a first reference signal and the carrier signal, and control the DC/AC inverter switches based on a comparison between a second reference signal and the carrier signal, wherein the at least one controller is further configured to control the DC/DC converter switches to provide power to the DC/AC inverter switches in synchronization with controlling the DC/AC inverter switches to draw power from the DC/DC converter switches.

2. The power-conversion system of claim 1, wherein the at least one controller is further configured to control at least two converter switches of the converter switches and at least two inverter switches of the inverter switches to be in a closed and conducting position approximately simultaneously.
3. The power-conversion system of claim 1, wherein the DC/DC converter includes a transformer having a primary winding coupled to the converter switches and a secondary winding coupled to the inverter switches.
4. The power-conversion system of claim 3, wherein the at least one controller is further configured to control at least two converter switches of the converter switches to conduct power from the input to the primary winding, and control at least two inverter switches of the inverter switches to conduct power from the secondary winding to the output.
5. The power-conversion system of claim 1, further comprising: a positive DC bus and a negative DC bus coupled between the DC/DC converter and the DC/AC inverter; and a capacitor coupled between the positive DC bus and the negative DC bus.
6. The power-conversion system of claim 5, wherein the at least one controller is further configured to control the DC/AC inverter to receive the converted power directly from the DC/DC converter, bypassing the capacitor.
7. The power-conversion system of claim 5, wherein the capacitor has a capacitance of approximately 50 μF or less.
8. The power-conversion system of claim 5, further comprising an output inductor coupled to the capacitor via the DC/AC inverter and coupled to the output.
9. The power-conversion system of claim 8, wherein the at least one controller is further configured to control the DC/AC inverter to discharge the output inductor to the capacitor.
10. The power-conversion system of claim 8, wherein the at least one controller is further configured to control the DC/AC inverter to discharge the capacitor to the output inductor.
11. The power-conversion system of claim 1, wherein the first reference signal and the second reference signal have a same magnitude and a same frequency, and are phase-shifted by 180 degrees with respect to one another.
12. A non-transitory computer-readable medium storing thereon sequences of computer-executable instructions for operating a power-conversion system having an input, an output, a DC/DC converter coupled to the input, a DC/AC inverter coupled to the DC/DC converter and to the output, wherein the DC/DC converter includes converter switches coupled to the input, and wherein the DC/AC inverter includes inverter switches coupled to the output, the sequences of computer-executable instructions including instructions that instruct at least one processor to: control the DC/DC converter based on a carrier signal having a carrier-signal frequency to draw input power from the input and provide converted power to the DC/AC inverter, control the DC/AC inverter to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input, wherein controlling the DC/AC inverter to provide the output power to the output in synchronization with the DC/DC converter drawing the input power from the input includes controlling the DC/AC inverter based on the carrier signal having the carrier-signal frequency, control the DC/DC converter switches based on a comparison between a first reference signal and the carrier signal, control the DC/AC inverter switches based on a comparison between a second reference signal and the carrier signal, and control the DC/DC converter switches to provide power to the DC/AC inverter switches in synchronization with controlling the DC/AC inverter switches to draw power from the DC/DC converter switches.
13. The non-transitory computer-readable medium of claim 12, wherein the DC/DC converter includes at least two converter switches and the DC/AC inverter includes at least two inverter switches, and wherein the instructions further instruct at least one processor to: control the at least two converter switches to provide power to the at least two inverter switches in synchronization

with controlling the at least two inverter switches to draw power from the at least two converter switches.

14. The non-transitory computer-readable medium of claim 13, wherein the instructions further instruct at least one processor to control the at least two converter switches and the at least two inverter switches to be in a closed and conducting position approximately simultaneously.

15. The non-transitory computer-readable medium of claim 13, wherein the DC/DC converter includes a transformer having a primary winding and a secondary winding, wherein controlling the at least two converter switches to provide power to the at least two inverter switches includes controlling the at least two converter switches to provide power to the primary winding, and wherein controlling the at least two inverter switches to receive the converted power and provide output power to the output in synchronization with the DC/DC converter drawing the input power from the input includes controlling the at least two inverter switches to conduct power from the secondary winding to the output.

16. The non-transitory computer-readable medium of claim 12, wherein the power-conversion system further includes a positive DC bus and a negative DC bus coupled between the DC/DC converter and the DC/AC inverter, and a capacitor coupled between the positive DC bus and negative DC bus, and wherein the instructions further instruct at least one processor to control the DC/AC inverter to receive the converted power directly from the DC/DC converter, bypassing the capacitor.

17. The non-transitory computer-readable medium of claim 12, wherein the power-conversion system further includes a capacitor coupled to the DC/AC inverter and an output inductor coupled to the capacitor via the DC/AC inverter and to the output, wherein the instructions further instruct at least one processor to control the DC/AC inverter to discharge the output inductor to the capacitor.

18. The non-transitory computer-readable medium of claim 12, wherein the power-conversion system further includes a capacitor coupled to the DC/AC inverter and an output inductor coupled to the capacitor via the DC/AC inverter and to the output, wherein the instructions further instruct at least one processor to control the DC/AC inverter to discharge the capacitor to the output inductor.

19. A method of operating a power-conversion system having an input, an output, a DC/DC converter coupled to the input, a DC/AC inverter coupled to the DC/DC converter and to the output, wherein the DC/DC converter includes converter switches coupled to the input, and wherein the DC/AC inverter includes inverter switches coupled to the output, the method comprising: controlling the DC/DC converter based on a carrier signal having a carrier-signal frequency to draw input power from the input and provide power to the DC/AC inverter; controlling the DC/AC inverter to receive the converted power and provide output power to the output in synchronization with the DC/AC inverter drawing power from the DC/DC converter, wherein controlling the DC/AC inverter to receive the converted power and provide the output power to the output in synchronization with the DC/AC inverter drawing power from the DC/DC converter includes controlling the DC/AC inverter based on the carrier signal having the carrier-signal frequency; controlling the DC/DC converter switches based on a comparison between a first reference signal and the carrier signal; and controlling the DC/AC inverter switches based on a comparison between a second reference signal and the carrier signal, wherein controlling the DC/DC converter switches to provide power to the DC/AC inverter switches is in synchronization with controlling the DC/AC inverter switches to draw power from the DC/DC converter switches.
